
WAEC Mathematics 2025 Past Questions

Problems generated from the Cheetah WAEC 2025 question bank.

Question 1 (MCQ)

Find $(101_2)^2$, expressing the answer in base 2.

- A. 11101
- B. 10101
- C. 10010
- D. 11001

Question 2 (MCQ)

If three children share N10.50 among themselves in the ratio 6:7:8, how much is the largest share?

- A. N3.50
- B. N4.00
- C. N4.50
- D. N3.00

Question 3 (MCQ)

Express 0.000834 in standard form.

- A. 8.34×10^{-4}
- B. 8.34×10^{-3}
- C. 8.34×10^4
- D. 8.34×10^3

Question 4 (MCQ)

Simplify $0.027^{-1/3}$.

- A. $3/10$
- B. $3 \frac{1}{3}$
- C. $1/3$
- D. 3

Question 5 (MCQ)

Given that $\log_2(a) = \log_8(4)$, find a.

- A. $2^{1/2}$
- B. $2^{2/3}$
- C. $4^{2/3}$
- D. $4^{1/2}$

Question 6 (MCQ)

By selling some crates of soft drinks for N600.00, a dealer makes a profit of 50%. How much did the dealer pay for the drinks?

- A. N400.00
- B. N1200.00
- C. N900.00
- D. N450.00

Question 7 (MCQ)

Find the n th term U_n of the arithmetic progression 11, 4, -3, ...

- A. $U_n = 19 - 7n$
- B. $U_n = 19 + 7n$
- C. $U_n = 18 - 7n$
- D. $U_n = 18 + 7n$

Question 8 (MCQ)

If $R = \{2, 4, 6, 7\}$ and $S = \{1, 2, 4, 8\}$, then R union S equals what?

- A. $\{1, 2, 4, 6, 7, 8\}$
- B. $\{1, 2, 4, 7, 8\}$
- C. $\{1, 4, 7, 8\}$
- D. $\{2, 6, 7\}$

Question 9 (MCQ)

If $16/9$, x , 1 , y are in geometric progression, find the product xy .

- A. $16/9$
- B. $3/4$
- C. 1
- D. $4/3$

Question 10 (MCQ)

[Diagram required] In the diagram, the shaded portion is represented by which set expression?

- A. $(P \cup Q) \cap R$
- B. $P \cap Q \cap R$
- C. $P \cap R$
- D. $Q \cap R$

Question 11 (MCQ)

Find the values of x for which $(6x - 1)/(x^2 + 4x - 5)$ is undefined.

- A. +5 or -1
- B. -5 or +1
- C. -5 or -1
- D. +4 or +1

Question 12 (MCQ)

Solve the inequality $(1/3)(2x - 1) < 5$.

- A. $x < -5$
- B. $x < 8$
- C. $x < -6$
- D. $x < 7$

Question 13 (MCQ)

Find the smaller value of x for which $x^2 - 3x + 2 = 0$.

- A. 4
- B. 2
- C. 3
- D. 1

Question 14 (MCQ)

Factorize $x(a - c) + y(c - a)$.

- A. $(a - c)(y - x)$
- B. $(a + c)(x + y)$
- C. $(a + c)(x - y)$
- D. $(a - c)(x - y)$

Question 15 (MCQ)

If y varies inversely as x^2 and $x = 3$ when $y = 4$, find y when $x = 2$.

- A. 3
- B. 9
- C. 1
- D. 18

Question 16 (MCQ)

Solve $3x^2 + 25x - 18 = 0$.

- A. -3, 2
- B. -9, $\frac{2}{3}$
- C. -2, 9
- D. -2, 3

Question 17 (MCQ)

Solve $(x + 2)(x - 7) = 0$.

- A. -1 or 8
- B. -4 or 5
- C. -3 or 6
- D. -2 or 7

Question 18 (MCQ)

Solve $x + y = \frac{3}{2}$ and $x - y = \frac{5}{2}$. Use your result to find $2y + x$.

- A. -2
- B. $\frac{1}{2}$
- C. 1
- D. -1

Question 19 (MCQ)

[Graph required] Which inequality is illustrated by the sketch graph? The boundary line passes through (0, 3) and (1, 0), with the region below the line shaded.

- A. $y \leq -3x + 3$
- B. $y \leq 3x + 2$
- C. $y \geq -2x + 3$
- D. $y \leq x + 3$

Question 20 (MCQ)

For the graph of $y = 3x^2 + x - 7$, what is the minimum value of y from the given options?

- A. -7
- B. -10
- C. -4
- D. -1

Question 21 (MCQ)

Using the graph of $y = 3x^2 + x - 7$, find the roots of the equation.

- A. 1.4 and -1.7
- B. 2.0 and -1.9
- C. 1.0 and -1.2
- D. 1.1 and -1.3

Question 22 (MCQ)

The diagonals AC and BD of a rhombus ABCD are 16 cm and 12 cm long. Calculate the area of the rhombus.

- A. 96 cm^2
- B. 36 cm^2
- C. 48 cm^2
- D. 24 cm^2

Question 23 (MCQ)

A cylindrical container, closed at both ends, has radius 7 cm and height 5 cm. Find its total surface area.

- A. 35 cm^2
- B. 528 cm^2
- C. 154 cm^2
- D. 220 cm^2

Question 24 (MCQ)

Find the total surface area of a solid circular cone with base radius 3 cm and slant height 4 cm. Take $\pi = \frac{22}{7}$.

- A. 66 cm^2
- B. $37 \frac{5}{7} \text{ cm}^2$
- C. $75 \frac{3}{7} \text{ cm}^2$
- D. $78 \frac{6}{7} \text{ cm}^2$

Question 25 (MCQ)

A water tank of height $\frac{1}{2}$ m has a square base of side $1 \frac{1}{2}$ m. It is filled from a water tanker holding 1500 litres. How many litres of water are left?

- A. 375 litres
- B. 3750 litres
- C. 37.5 litres
- D. 37500 litres

Question 26 (MCQ)

A sphere has volume $k \text{ cm}^3$ and surface area $k \text{ cm}^2$. Calculate the diameter of the sphere.

- A. 3 cm
- B. 9 cm
- C. More information needed
- D. 6 cm

Question 27 (MCQ)

[Diagram required] A cone has a frustum whose indicated radii are 12 cm and 6 cm and whose frustum slant height is 12 cm. Calculate the full cone slant height.

- A. 24 cm
- B. 15 cm
- C. 8 cm
- D. 12 cm

Question 28 (MCQ)

The positions of two countries P and Q are (15 degrees N, 12 degrees E) and (65 degrees N, 12 degrees E). What is the difference in latitude?

- A. 50 degrees
- B. 104 degrees
- C. 80 degrees
- D. 100 degrees

Question 29 (MCQ)

A 120 degree sector of a circle of radius 21 cm is bent to form a cone. What is the base radius of the cone?

- A. $3\frac{1}{2}$ cm
- B. $10\frac{1}{2}$ cm
- C. 14 cm
- D. 7 cm

Question 30 (MCQ)

A cylindrical container, closed at both ends, has radius 7 cm and height 5 cm. What is the volume of the container?

- A. 770 cm^3
- B. 35 cm^3
- C. 220 cm^3
- D. 154 cm^3

Question 31 (MCQ)

If $\log_{10}(x) = 2.3675$ and $\log_{10}(y) = 2.9738$, find $x + y$, correct to three significant figures.

- A. 0.118
- B. 0.903
- C. 0.117
- D. 0.944

Question 32 (MCQ)

A sector of a circle has angle 108 degrees and radius $3\frac{1}{2}$ cm. Find the perimeter of the sector.

- A. $6\frac{3}{5}$ cm
- B. $7\frac{2}{10}$ cm
- C. $6\frac{4}{5}$ cm
- D. $13\frac{3}{5}$ cm

Question 33 (MCQ)

[Diagram required] In the diagram, AO is perpendicular to OB. Find x.

- A. 15 degrees
- B. 22.5 degrees
- C. 7.5 degrees
- D. 30 degrees

Question 34 (MCQ)

[Diagram required] In the diagram, O is the centre of the circle and $BD = DC$. If angle $DCB = 35$ degrees, find angle BAO.

- A. 25 degrees
- B. 30 degrees
- C. 20 degrees
- D. 35 degrees

Question 35 (MCQ)

Which of the following angles is an exterior angle of a regular polygon?

- A. 95 degrees
- B. 78 degrees
- C. 72 degrees
- D. 85 degrees

Question 36 (MCQ)

The locus of a point equidistant from two fixed points is the:

- A. angle bisector of the straight lines joining them
- B. parallel line to the straight line joining them
- C. perpendicular to the straight line joining them
- D. perpendicular bisector of the straight line joining them

Question 37 (MCQ)

[Diagram required] In the diagram, $AB = 12$ cm, $AE = 8$ cm, $DC = 9$ cm and the indicated lines are parallel. Find EC .

- A. 6 cm
- B. 5 cm
- C. 8 cm
- D. 9 cm

Question 38 (MCQ)

[Diagram required] In the diagram, PQ is parallel to TU , angle $PQR = 50$ degrees, angle $QRS = 86$ degrees and angle $STU = 64$ degrees. Calculate x .

- A. 100 degrees
- B. 108 degrees
- C. 120 degrees
- D. 136 degrees

Question 39 (MCQ)

[Diagram required] In the diagram, AB is parallel to DC , and the bisectors of angle BAC and angle ACD meet at E . Find angle AEC .

- A. 45 degrees
- B. 60 degrees
- C. 30 degrees
- D. 90 degrees

Question 40 (MCQ)

[Diagram required] In the diagram, AB is parallel to CD . What is the size of angle x ?

- A. 93 degrees
- B. 77 degrees
- C. 103 degrees
- D. 62 degrees

Question 41 (MCQ)

[Diagram required] In the diagram, find $b + c$.

- A. a degrees
- B. 90 degrees
- C. 180 degrees
- D. d degrees

Question 42 (MCQ)

In triangle ABC, side BC is produced to D. If $AB = AC$ and angle $BAC = 50$ degrees, find angle ACD.

- A. 115 degrees
- B. 65 degrees
- C. 60 degrees
- D. 50 degrees

Question 43 (MCQ)

In right triangle PQR, angle $PRQ = 90$ degrees, angle $QPR = 30$ degrees and $PQ = 10$ cm. Find QR.

- A. 6 cm
- B. 4 cm
- C. 5 cm
- D. 3 cm

Question 44 (MCQ)

What is the mode of 8, 10, 9, 9, 10, 8, 11, 8, 10, 9, 8 and 14?

- A. 8
- B. 9
- C. 10
- D. 11

Question 45 (MCQ)

The bearings of Q and R from P are 030 degrees and 120 degrees respectively. If $PQ = 12$ m and $PR = 5$ m, find QR.

- A. 11 m
- B. 13 m
- C. 9 m
- D. 7 m

Question 46 (MCQ)

Without using tables, evaluate $\sin 20^\circ / \cos 70^\circ + \cos 25^\circ / \sin 65^\circ$.

- A. 2
- B. 0
- C. 1
- D. -1

Question 47 (MCQ)

The mean of 20 observations is 4. If the largest observation is 23, find the mean of the remaining observations.

- A. 4
- B. 2.60
- C. 2.85
- D. 3

Question 48 (MCQ)

Two fair dice are tossed together once. Find the probability that the sum of the outcomes is at least 10.

- A. $\frac{1}{6}$
- B. $\frac{1}{12}$
- C. $\frac{1}{4}$
- D. $\frac{5}{36}$

Question 49 (MCQ)

Find the median of 2.64, 2.50, 2.72, 2.91 and 2.35.

- A. 2.50
- B. 2.64
- C. 2.91
- D. 2.72

Question 50 (MCQ)

A box contains 2 red, 6 white and 5 black balls. A ball is selected at random. What is the probability that it is black?

- A. $\frac{5}{13}$
- B. $\frac{5}{11}$
- C. $\frac{2}{13}$
- D. $\frac{5}{6}$

Question 51 (FRQ)

PART ONE. (a) Factorize $px^2 - 2qx - 4qy + 2py$. (b) Given $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $P = \{1, 2, 4, 6, 10\}$ and $Q = \{2, 3, 6, 9\}$, show that $(P \cup Q)^1 = P^1 \cap Q^1$.

Write your answer clearly, showing all necessary working.

Question 52 (FRQ)

(a) y is partly constant and partly varies inversely as the square of x . Write the relationship between x and y . (b) When $x = 1$, $y = 11$, and when $x = 2$, $y = 5$. Find y when $x = 4$.

Write your answer clearly, showing all necessary working.

Question 53 (FRQ)

(a) In the diagram, PQSR and SRYZ are parallelograms, and PQYZ is a straight line. If $QY = 2$ cm and $RS = 3$ cm, find PZ. (b) P and Q are towns on latitude 56 degrees N with longitudes 25 degrees E and 95 degrees E. Find distance PQ along the parallel of latitude, correct to nearest km. Take $R = 6400$ km and $\pi = 22/7$.

Write your answer clearly, showing all necessary working.

Question 54 (FRQ)

(a) A pack of 52 playing cards is shuffled and a card is drawn at random. Calculate the probability that it is either a five or a red nine. (b) P, Q and R are points in the same horizontal plane. Bearing of Q from P is 150 degrees and bearing of R from Q is 060 degrees. If $PQ = 5$ m and $QR = 3$ m, find the bearing of R from P, correct to nearest degree.

Write your answer clearly, showing all necessary working.

Question 55 (FRQ)

The table shows marks scored by 32 students. Marks: 1,2,3,4,5,6,7,8,9,10. Frequencies: 2,3,4,4,4,4,5,3,2,1. (a)(i) Find the mean. (ii) Find the median. (iii) Find the mode. (b) Find the percentage who scored at least 8 marks.

Write your answer clearly, showing all necessary working.

Question 56 (FRQ)

PART TWO. (a)(i) Using mathematical tables, find $2\sin 63.35$ degrees. (ii) Find $\log \cos 44.74$ degrees. (iii) Find k given that $\log k - \log(k - 2) = \log 5$. (b) Use logarithm tables to evaluate $(3.68^2 \times 6.705)/\sqrt{0.3581}$.

Write your answer clearly, showing all necessary working.

Question 57 (FRQ)

(a) Given $p = x + ym^3$, find m in terms of p , x and y . (b) By completing the square, find the roots of $x^2 - 6x + 7 = 0$ to 1 decimal place. (c) The product of two consecutive positive odd numbers is 195. Form and solve a quadratic equation to find the numbers.

Write your answer clearly, showing all necessary working.

Question 58 (FRQ)

(a) Complete the table for $y = 2\cos(2x) - 1$ for $x = 0, 30, 60, 90, 120, 150, 180$ degrees. (b) Draw the graph for $0 \leq x \leq 180$. (c) On the same axes, draw $y = (1/180)(x - 360)$. (d) Use the graphs to find x for which $2\cos(2x) + 1/2 = 0$.

Write your answer clearly, showing all necessary working.

Question 59 (FRQ)

(a) Using ruler and compasses only, construct triangle ABC with $AB = 7.5$ cm, $BC = 8.1$ cm and angle $ABC = 105$ degrees. (b) Locate D on BC such that $BD:DC = 3:2$. (c) Through D, construct a line perpendicular to BC. (d) If the line meets AC at P, measure BP.

Write your answer clearly, showing all necessary working.

Question 60 (FRQ)

(a)(i) A man travels from X on a bearing 060 degrees to Y, 20 km away. From Y he travels to Z on bearing 195 degrees. If Z is directly east of X, find YZ to 3 s.f. (ii) Find ZX to 3 s.f. (b)(i) An aircraft flies due south from latitude 36 degrees N to 36 degrees S on the same longitude. Find the distance travelled to 3 s.f. (ii) If its speed is 800 km/h, find the time to nearest hour. Take $\pi = \frac{22}{7}$, $R = 6400$ km.

Write your answer clearly, showing all necessary working.

Question 61 (FRQ)

The table shows scores of 2000 candidates: percent marks 11-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80, 81-90 with frequencies 68,184,294,402,480,310,164,98. Prepare a cumulative frequency table and draw the cumulative frequency curve.

Write your answer clearly, showing all necessary working.

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